College of Engineering – Department of Civil Engineering

CURRICULUM

Bachelor of Science in Civil Engineering

Academic Year 2025-2026

Reference CMOs: CMO No. 92 2017, CMO No. 20 s. 2013, CMO No. 39 s. 2021, CMO No. 40 s. 2021

Curriculum Description

Civil Engineering is a profession that applies the basic principles of science in conjunction with mathematical and computational tools to solve problems associated with developing and sustaining civilized life on our planet.

The Civil Engineering curriculum is designed to prepare graduates to apply knowledge of mathematics, calculus-based physics, chemistry, and at least one additional area of basic science, consistent with the Program Educational Objectives; apply knowledge of technical areas appropriate to civil engineering; conduct civil engineering experiments and analyze and interpret the resulting data; design a system component, or process in more than one civil engineering context; explain basic concepts in management, business, public policy, and leadership; and explain the importance of professional licensure.

Program Education Objectives

The civil engineering alumni three to five years after graduation shall:

PEO 1: Specialist

Practice as civil engineering specialist in structural, geotechnical, water resources, and transportation engineering, as well as construction engineering management for building the nation and betterment of society with due consideration for both the technical and also for the environmental, economic, social and other human concerns.

PEO 2: Professional and Leadership

Assumed leadership position in industry, academe, government, or private sector within consideration to social and ethical responsibility.

PEO 3: Lifelong Learning

Engaged in lifelong learning through further studies, research, certifications, promotions, and other personal and professional development activities.

Student Outcomes

The following skills, knowledge, and values are expected to be attained by students as they progress through the program:

SO1 Discipline Knowledge. Ability to apply mathematics, sciences and principles of engineering to solve complex civil engineering problems;

SO2 Investigation. Ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions;

SO3 Design/Development of Solutions. Design solution, system, components, processes, exhibiting improvements/innovations, that meet specified needs with appropriate consideration for public health and safety, cultural, societal, economical, ethical, environmental and sustainability issues.

SO4 Leadership and Teamwork. Function effectively as a member of a leader on a diverse team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.

SO5 Problem Analysis. Identify, formulate, and solve complex civil engineering problems by applying principles of engineering, science, and mathematics;

SO6 Ethics and Professionalism. Apply ethical principles and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of

engineering solutions in global, environmental, and societal contexts.

SO7 Communication. Communicate effectively on complex civil engineering activities with the community, and the society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions;

SO8 Environment and Sustainability. Recognize the impact of professional engineering solutions in societal, global, and environmental contexts and demonstrate knowledge of and need for sustainable development;

SO9 Lifelong Learning. Recognize the need for, and ability to engage in independent and life-long learning in the broadest context of technological change.

SO10 The Engineer and Society. Apply reasoning based on contextual knowledge to assess societal, health, safety, legal, cultural, contemporary issues, and the consequent responsibilities relevant to professional engineering practices.

SO11 Modern Tool Usage. Apply appropriate techniques, skills, and modern engineering and IT tools to complex civil engineering activities;

SO12 Project Management and Finance. Demonstrate knowledge and understanding of engineering management and financial principles as member or a leader of a team to manage projects in multidisciplinary settings and identify opportunities of entrepreneurship.

SO13 Social and National Responsibility. Apply acquired engineering knowledge and skills in addressing community problems that contributes to national development.

CURRICULUM COMPONENTS

Classification / Field / Course	Number of Hours per Week		Credit Units
	Lec	Lab	
I. TECHNICAL COURSES			
A. Mathematics			
Differential Calculus	4	0	4
Integral Calculus	4	0	4
Differential Equations	3	0	3
Probability and Statistics for AI	3	0	3
Numerical Solutions to CE Problems	2	3	3
Subtotal	***16***	***3***	***17***
B. Natural/ Physical Sciences			
Chemistry for Engineers	3	3	4
Physics for Engineers 1	3	3	4
Physics for Engineers 2	2	3	3
Geology	2	0	2
Environmental Science and Engineering	3	0	3
Subtotal	***13***	***9***	***16***
C. Basic Engineering Sciences			
Civil Engineering Orientation	2	0	2
Introduction to Engineering	0	3	1
Engineering Drawing	0	3	1
Computer Programming 1	0	3	1
Computer Programming 2	0	3	1
Computer-Aided Design	0	3	1
Statics of Rigid Bodies	3	0	3
Dynamics of Rigid Bodies	2	0	2
Mechanics of Deformable Bodies	4	0	4
Engineering Economics	3	0	3
Engineering Management	2	0	2
Technopreneurship	3	0	3
Basic Occupational Safety and Health	0	3	1
Subtotal	***19***	***18***	***25***

College of Engineering – Department of Civil Engineering

D. Allied Courses			
Engineering Utilities 1	3	0	3
Engineering Utilities 2	3	0	3
Subtotal	***6***	***0***	***6***
E. Professional Engineering Courses			
Fundamentals of Surveying	3	6	5
Construction Materials and Testing	2	3	3
Structural Theory	3	3	4
Principles of Reinforced/ Prestressed Concrete Design	3	3	4
Hydraulics	3	0	3
Hydraulics Laboratory	1	3	2
Hydrology	3	0	3
CE Laws, Ethics and Contracts Specifications	2	0	2
Highway and Railroad Engineering	3	0	3
Building System Design	2	3	3
Geotechnical Engineering 1: Soil Mechanics	3	3	4
Principles of Transportation Engineering	3	0	3
Quantity Surveying	1	3	2
Construction Methods and Project Management	3	0	3
Principles of Steel Design	2	3	3
GIS Application for Engineering Surveys	0	3	1
CE Capstone Design Project 1	1	3	2
CE Capstone Design Project 2	0	6	2
CE Practice with Comprehensive Examination	0	6	2
Methods of Research	3	0	3
OJT/Internship (320 hrs)		320 hours	4
Subtotal	***41***	***48***	***61***
F. Professional Courses – Specialized			
Professional Course- specialized 1	3	0	3
Professional Course- specialized 2	3	0	3
Professional Course- specialized 3	3	0	3
Professional Course- specialized 4	3	0	3
Professional Course- specialized 5	3	0	3
Subtotal	***15***	***0***	***15***
TOTAL TECHNICAL COURSES	**110**	**78**	**140**
II. NON-TECHNICAL COURSES			
A. General Education Courses			
Understanding the Self	3	0	3
Mathematics in the Modern World	3	0	3
The Contemporary World	3	0	3
Readings in Philippine History	3	0	3
Purposive Communication	3	0	3
Ethics	3	0	3
Art Appreciation	3	0	3
Science, Technology and Society	3	0	3
Subtotal	***24***	***0***	***24***
B. General Education Electives and Mandated Course			
Linear Algebra for AI	3	0	3
Machine Learning and Neural Networks	3	0	3
The Entrepreneurial Mind	3	0	3
Life and Works of Rizal	3	0	3
Subtotal	***12***	***0***	***12***
C. Physical Activities Toward Health and Fitness (PATHFit)			
Movement Competency Training	2	0	2
Exercise-based Fitness Activities	2	0	2

College of Engineering – Department of Civil Engineering

Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities 1	2	0	2
Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities 2	2	0	2
Subtotal	***8***	***0***	***8***
D. National Service Training Program			
National Service Training Program 1	3	0	3
National Service Training Program 2	3	0	3
Subtotal	***6***	***0***	***6***
TOTAL NON-TECHNICAL COURSES	**50**	**0**	**50**
GRAND TOTAL	**160**	**78**	**190**

SUMMARY

Courses	Number of Units
I. TECHNICAL COURSES	
A. Mathematics	17
B. Natural/Physical Sciences	16
C. Basic Engineering Sciences	25
D. Allied Courses	6
E. Professional Engineering Courses	61
F. Professional – Specialized Courses	15
II. NON-TECHNICAL COURSES	
A. Required General Education Courses	24
B. General Education Elective and Mandated Courses	12
C. Physical Activities Toward Health and Fitness (PATHFit)	8
D. National Service Training Program	6
GRAND TOTAL	**190**

CURRICULUM STRUCTURE

FIRST YEAR						
FIRST SEMESTER						
Course Code	**Course Title**	**Credit Units**	**Lec**	**Lab**	**Prerequisite(s)**	**Co-requisite(s)**
CHEM 101	Chemistry for Engineers	4	3	3		
GEd 105	Readings in Philippine History	3	3	0		
GEd 108	Art Appreciation	3	3	0		
MATH 101	Differential Calculus	4	4	0		
GEd 104	The Contemporary World	3	3	0		
NSTP 111	National Service Training Program 1	3	3	0		
PATHFit 1	Movement Competency Training	2	2	0		
CpE 102	Computer Programming 1	1	0	3		
ENGG 101	Introduction to Engineering	1	0	3		
	TOTAL	***24***	***21***	***9***		
FIRST YEAR						
SECOND SEMESTER						
Course Code	**Course Title**	**Credit Units**	**Lec**	**Lab**	**Prerequisite(s)**	**Co-requisite(s)**

College of Engineering – Department of Civil Engineering

PHYS 102	Physics for Engineers 1	4	3	3	MATH 101	
GEd 102	Mathematics in the Modern World	3	3	0		
GEd 106	Purposive Communication	3	3	0		
GEd 101	Understanding the Self	3	3	0		
MATH 102	Integral Calculus	4	4	0	MATH 101	
NSTP 121	National Service Training Program 2	3	3	0	NSTP 111	
PATHFit 2	Exercise-based Fitness Activities	2	2	0	PATHFit 1	
ENGG 102	Engineering Drawing	1	0	3		
GEd 109	Science, Technology and Society	3	3	0		
	TOTAL	***26***	***24***	***6***		
FIRST YEAR						
MIDTERM SEMESTER						
Course Code	**Course Title**	**Credit Units**	**Lec**	**Lab**	**Prerequisite(s)**	**Co-requisite(s)**
PHYS 103	Physics for Engineers 2	3	2	3	PHYS 102	
GEd 107	Ethics	3	3	0		
GEd 103	Life and Works of Rizal	3	3	0		
	TOTAL	***9***	***8***	***3***		
SECOND YEAR						
FIRST SEMESTER						
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	
MATH 103	Differential Equations	3	3	0	MATH 102	
ENGG 107	Statics of Rigid Bodies	3	3	0	PHYS 102	
CE 201	Civil Engineering Orientation	2	2	0		
ME 209	Engineering Utilities 2	3	3	0	PHYS 103	
EE 209	Engineering Utilities 1	3	3	0	PHYS 103	
ENGG 103	Computer-Aided Design	1	0	3	ENGG 102	
CpE 103	Computer Programming 2	1	0	3	CpE 102	
PATHFit 3	Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities 1	2	2	0	PATHFit 1 & 2	
AI 101	Linear Algebra for AI	3	3	0		
GEdE 101	The Entrepreneurial Mind	3	3	0		
ENGG 108	Dynamics of Rigid Bodies	2	2	0	PHYS 102	
GeoE 200	Geology	2	2	0		
	TOTAL	***28***	***26***	***6***		

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College of Engineering – Department of Civil Engineering

SECOND YEAR						
SECOND SEMESTER						
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**
ENGG 111	Mechanics of Deformable Bodies	4	4	0	ENGG 107	
CE 202	Construction Materials and Testing	3	2	3		ENG 111
CE 203	Fundamentals of Surveying	5	3	6		
AI 102	Probability and Statistics for AI	3	3	0	MATH 102	
CE 204	Building System Design	3	2	3	ENGG 102	
PATHFit 4	Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities 2	2	2	0	PATHFit 1 & 2	
CE 205	Hydraulics	3	3	0	ENGG 107, ENGG 108	ENGG 111
AI 103	Machine Learning and Neural Networks	3	3	0		
	TOTAL	***26***	***22***	***12***		
THIRD YEAR						
FIRST SEMESTER						
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	
CE 301	Structural Theory	4	3	3	ENGG 111	
CE 302	Hydraulics Laboratory	2	1	3	CE 205	
CE 303	Quantity Surveying	2	1	3	CE 204	
CE 304	GIS Application for Engineering Surveys	1	0	3	CE 203	
CE 305	Highway and Railroad Engineering	3	3	0	CE 203	
CE 306	Geotechnical Engineering 1: Soil Mechanics	4	3	3	CE 205 GeoE 200	
CE 307	CE Laws, Ethics and Contracts Specifications	2	2	0		
ENGG 116	Methods of Research	3	3	0	AI 102. 3rd year standing	
CE 308	Hydrology	3	3	0		
	TOTAL	***24***	***19***	***15***		
THIRD YEAR						
SECOND SEMESTER						
Course Code	**Course Title**	**Credit Units**	**Lec**	**Lab**	**Prerequisite(s)**	**Co-requisite(s)**
CE 309	Principles of Reinforced/ Prestressed Concrete Design	4	3	3	CE 301	
CE 310	Construction Methods and Project Management	3	3	0	CE 204 CE 303	

College of Engineering – Department of Civil Engineering

CE 311	Principles of Steel Design	3	2	3	CE 301	
CE 312	Principles of Transportation Engineering	3	3	0	CE 305	
CE 313	Numerical Solutions to CE Problems	3	2	3	MATH 201	
	Professional Course-specialized 1	3	3	0		
	Professional Course - specialized 2	3	3	0		
CE 314	CE Capstone Design Project 1	2	1	3	ENGG 116 3rd Standing	
	TOTAL	***24***	***20***	***12***		
THIRD YEAR						
MIDTERM SEMESTER						
Course Code	**Course Title**	**Credit Units**	**Lec**	**Lab**	**Prerequisite(s)**	**Co-requisite(s)**
ENGG 114	Basic Occupational Safety and Health	1	0	3		
ENGG 106	Engineering Management	2	2	0	3rd year standing	
ENGG 104	Engineering Economics	3	3	0	MATH 101	
	TOTAL	***6***	***5***	***3***		
FOURTH YEAR						
FIRST SEMESTER						
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**
ENGG 117	OJT/Internship (320 hrs)	4		320 HRS	4th year standing	
CE 401	CE Capstone Design Project 2	2	0	6	CE 314	
	TOTAL	***6***	***0***	***6***		
FOURTH YEAR						
SECOND SEMESTER						
Course Code	**Course Title**	**Credit Units**	**Lec**	**Lab**	**Prerequisite(s)**	**Co-requisite(s)**
ENGG 113	Environmental Science and Engineering	3	3	0		
	Professional Course - specialized 3	3	3	0		
	Professional Course - specialized 4	3	3	0		
	Professional Course - specialized 5	3	3	0		
ENGG 105	Technopreneurship	3	3	0	4th year standing	
CE 402	CE Practice with Comprehensive Examination	2	0	6	Graduating	
	TOTAL	***17***	***15***	***6***		
GRAND TOTAL		**190**	**160**	**78**		

College of Engineering – Department of Civil Engineering

Professional Courses- Specialized

A. Construction Engineering and Management

Course Code	Course Title	Credit Units	Lec	Lab	Prerequisite(s)	Co-requisite(s)
CEM 401	Professional Course- specialized 1 - Computer Softwares for Construction Management	3	0	3		CE 310
CEM 402	Professional Course- specialized 2 -Advanced Construction Methods and Equipment	3	0	3		CE 310
CEM 403	Professional Course- specialized 3 -Construction Cost Engineering	3	0	3	CE 303	
CEM 404	Professional Course- specialized 4- Database Management in Construction	3	0	3	CE 303	
CEM 405	Professional Course- specialized 5- Construction Occupational Safety and Health	3	0	3	ENGG 114	

B. Geotechnical Engineering

Course Code	Course Title	Credit Units	Lec	Lab	Prerequisite(s)	Co-requisite(s)
CEG 401	Professional Course- specialized 1- Geotechnical Engineering 2: Rock Mechanics	3	0	3	CE 306	
CEG 402	Professional Course- specialized 2- Foundation Engineering	3	0	3	CE 306	
CEG 403	Professional Course- specialized 3 -Geotechnical Earthquake Engineering	3	0	3	CE 306	
CEG 404	Professional Course- specialized 4- Ground Improvement	3	0	3	CE 306	
CEG 405	Professional Course- specialized 5- Computer Softwares in Geotechnical Engineering	3	0	3	CE 306	

College of Engineering – Department of Civil Engineering

C. Structural Engineering

Course Code	Course Title	Credit Units	Lec	Lab	Prerequisite(s)	Co-requisite(s)
CES 401	Professional Course- specialized 1- Computer Softwares in Structural Analysis	3	0	3	CE 301	
CES 402	Professional Course- specialized 2- Earthquake Engineering	3	0	3	CE 301	
CES 403	Professional Course- specialized 3 -Foundation and Retaining Wall Design	3	0	3	CE 301 CE 306	
CES 404	Professional Course- specialized 4- Design of Steel Structure	3	0	3	CE 311	
CES 405	Professional Course- specialized 5- Bridge Engineering	3	0	3	CE 301 CE 306	

D. Transportation Engineering

Course Code	Course Title	Credit Units	Lec	Lab	Prerequisite(s)	Co-requisite(s)
CET 401	Professional Course- specialized 1- Computer Softwares for Transportation Engineering	3	0	3	CE 305	
CET 402	Professional Course- specialized 2 - Pavement Systems Design Engineering	3	0	3	CE 305 CE 310	
CET 403	Professional Course- specialized 3 -Airport Design	3	0	3	CE 309	
CET 404	Professional Course- specialized 4- Ports and Harbor Design	3	0	3	CE 205 CE 305	
CET 405	Professional Course- specialized 5- Railway Design Engineering	3	0	3	CE 305	

College of Engineering – Department of Civil Engineering

E. Water Resources Engineering

Course Code	Course Title	Credit Units	Lec	Lab	Prerequisite(s)	Co-requisite(s)
CEW 401	Professional Course- specialized 1- Water Resources Engineering	3	0	3	CE 205 CE 308	
CEW 402	Professional Course- specialized 2- Water Supply Planning and Development	3	0	3	CE 205	
CEW 403	Professional Course- specialized 3 -Irrigation Engineering	3	0	3	CE 205 CE 308	
CEW 404	Professional Course- specialized 4- Flood Control and Drainage Design	3	0	3	CE 205 CEW 401	
CEW 405	Professional Course- specialized 5- River Engineering	3	0	3	CE 205 CE 308	